

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-09 22:30 UTC

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Generated 2026-05-09 22:30 UTC

- Cycles recorded: **508**
- Time span: 384.8h (~1.32 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	473
BARRIER_HIT	16
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	25
Free Probability	12
Schur-Weyl Duality	11
Matroid Theory	9
Representation Theory of Symmetric Groups	9
Algebraic Geometry	7
Random Matrix Theory	6
Additive Combinatorics	6
Spectral Graph Theory	6
Polymatroid Theory	6
Finite Geometry	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
FOURIER_ANALYSIS	4
Plethysm Theory	4
Fourier analysis of boolean functions	3
Algebraic Topology	3

Field	Cycles
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3
Persistent Homology	3
Noncommutative geometry	2
Representation theory of symmetric groups	2
Ergodic Circuit Framework (COMM_COMPLEXITY)	2
Tropical geometry	2
Schur-Weyl duality, plethysm, algebraic combinatorics	2

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-09 18:32 UTC	INCONCLUSIVE	Genus Inverse Proportionality to Disjointness Communication Compl
2026-05-09 18:58 UTC	INCONCLUSIVE	Free Entropy Lower Bound on Disjointness Communication Complexity
2026-05-09 19:24 UTC	INCONCLUSIVE	Symmetric Group Irreducible Component Exponential Gap in Permanen
2026-05-09 19:47 UTC	INCONCLUSIVE	Quadratic Form Rank Inverse Proportional to SOS Refutation Degree
2026-05-09 19:56 UTC	INCONCLUSIVE	Algebraic Circuit Complexity via Monodromy Action on Tseitin Form
2026-05-09 20:05 UTC	INCONCLUSIVE	Newton Polytope Vertex Count Inverse Proportional to SOS Degree f
2026-05-09 20:11 UTC	INCONCLUSIVE	Real Stable Polynomial Degree Lower Bound for Max-CUT SOS
2026-05-09 20:27 UTC	INCONCLUSIVE	Young Tableau Count Inverse Proportional to Monotone Circuit Size
2026-05-09 21:07 UTC	INCONCLUSIVE	Real Root Count Inverse Proportional to SOS Degree for Max-CUT
2026-05-09 21:15 UTC	INCONCLUSIVE	Betti Number Inverse Proportionality to DPLL Tree Size in 3-SAT

Time	Verdict	Title
2026-05-09 21:30 UTC	INCONCLUSIVE	Schur-Weyl Decomposition Irreducible Component Gap for Permanent
2026-05-09 21:52 UTC	INCONCLUSIVE	Cheeger Constant Exponentiates Resolution Length for Tseitin Form
2026-05-09 22:03 UTC	INCONCLUSIVE	p-adic Valuation of Solution Count Modulo p Inversely Proportiona
2026-05-09 22:15 UTC	INCONCLUSIVE	Slice Rank Lower Bound for Disjointness Communication Matrices
2026-05-09 22:28 UTC	INCONCLUSIVE	Symmetric Tensor Rank Gap in Permanent vs Determinant Decompositi

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```